

Step 4: Interaction Discovery After Cleaning

Step 4 tests interpretable interaction terms after the Step 3 family-cleaned model. The candidate pool is broader than the exact Step 3 model: it includes selected variables plus reasonable non-duplicate family alternatives. Variables removed as exact duplicates, inverse codings, or near-compositional redundancies are not reintroduced.

Baseline vs Interaction-Augmented PLS

Model	Predictors	Components	CV R2	CV RMSE	Train R2
Step 3 cleaned PLS	27	5	0.558	0.058	0.625
Step 4 interaction PLS	38	1	0.566	0.057	0.587

Screening Rule

Each candidate interaction was tested by adding the interaction plus its main effects if those main effects were not already in the cleaned model. The screen used cross-validated PLS performance and the interaction term's VIP. I treated an interaction as promising only when it did not worsen CV RMSE and had interaction VIP around 0.8 or higher.

Best Interaction Trials

Interaction	Families	CV R2	CV RMSE	VIP	Story
block1_low_income_lim_at_share__x_block5_requests_311_per_1000	household_class + service_contact	0.618	0.054	0.954	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block1_bachelors_or_higher_25_64_share__x_block5_requests_311_per_1000	household_class + service_contact	0.607	0.055	0.775	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block5_ksi_collision_events_2021_2025_per_1000__x_block1_unemployment_rate_share	service_contact + household_class	0.596	0.055	0.808	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block1_low_income_lim_at_share__x_block5_development_applications_2021_2025_per_1000	household_class + service_contact	0.595	0.055	0.645	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block1_average_household_size__x_block5_requests_311_per_1000	household_class + service_contact	0.593	0.055	0.711	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.

Interaction	Families	CV R2	CV RMSE	VIP	Story
block5_ksi_collision_events_2021_2025_per_1000__x_block3_non_official_mother_tongue_share	service_contact + immigration_citizenship	0.591	0.056	1.086	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.
block1_bachelors_or_higher_25_64_share__x_block1_median_age	household_class + age_structure	0.584	0.056	0.774	Tests whether the class/household-size turnout pattern changes across age structure.
block5_requests_311_per_1000__x_block3_immigrant_share	service_contact + immigration_citizenship	0.582	0.056	0.651	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.
block1_low_income_lim_at_share__x_block1_age_35_64_share	household_class + age_structure	0.582	0.056	0.933	Tests whether the class/household-size turnout pattern changes across age structure.
block1_average_household_size__x_block5_ksi_collision_events_2021_2025_per_1000	household_class + service_contact	0.581	0.056	0.827	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block5_transit_commute_share_preferred__x_block5_requests_311_2023_2025_estimated_count	transportation_accesses + service_contact	0.580	0.056	0.593	Theory-screened cross-family interaction with interpretable main effects retained.
block5_development_applications_2021_2025_per_1000__x_block3_immigrant_share	service_contact + immigration_citizenship	0.580	0.056	0.613	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.
block5_requests_311_per_1000__x_block3_non_official_mother_tongue_share	service_contact + immigration_citizenship	0.578	0.057	0.623	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.
block1_bachelors_or_higher_25_64_share__x_block5_requests_311_2023_2025_estimated_count	household_class + service_contact	0.577	0.057	0.618	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block1_bachelors_or_higher_25_64_share__x_block2_renter_share	household_class + housing_tenure	0.577	0.057	0.541	Theory-screened cross-family interaction with interpretable main effects retained.

Interaction	Families	CV R2	CV RMSE	VIP	Story
block3_non_citizen_share__x_block5_development_applications_2021_2025_per_1000	immigration_citizenship + service_contact	0.577	0.057	0.403	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.
block1_age_65_plus_share__x_block1_average_household_size	age_structure + household_class	0.576	0.057	0.950	Tests whether the class/household-size turnout pattern changes across age structure.
block1_unemployment_rate_share__x_block5_requests_311_2023_2025_estimated_count	household_class + service_contact	0.576	0.057	0.629	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block3_visible_minority_share__x_block5_development_applications_2021_2025_per_1000	immigration_citizenship + service_contact	0.576	0.057	0.566	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.
block1_low_income_lim_at_share__x_block5_requests_311_2023_2025_estimated_count	household_class + service_contact	0.574	0.057	0.395	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block1_average_household_size__x_block5_development_applications_2021_2025_per_1000	household_class + service_contact	0.573	0.057	0.282	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block1_bachelors_or_higher_25_64_share__x_block5_development_applications_2021_2025_per_1000	household_class + service_contact	0.572	0.057	0.605	Tests whether service-contact intensity has a different turnout meaning in higher-need or higher-resource areas.
block1_age_65_plus_share__x_block1_bachelors_or_higher_25_64_share	age_structure + household_class	0.572	0.057	0.690	Tests whether the class/household-size turnout pattern changes across age structure.
block1_low_income_lim_at_share__x_block2_condo_share	household_class + housing_form_density	0.571	0.057	0.773	Theory-screened cross-family interaction with interpretable main effects retained.
block3_recent_immigrant_share__x_block5_development_applications_2021_2025_per_1000	immigration_citizenship + service_contact	0.571	0.057	0.352	Tests whether service-contact intensity interacts with immigrant, language, or citizenship geography.

Interactions Retained In The Augmented PLS

Interaction	VIF	Turnout corr	Direction
block1_bachelors_or_higher_25_64_share_x_block4_effective_mayoral_candidates_5pct	0.863	0.240	higher turnout
block5_ksi_collision_events_2021_2025_per_1000_x_block3_non_official_mother_tongue_share	0.846	0.235	higher turnout
block1_low_income_lim_at_share_x_block1_age_35_64_share	0.798	-0.222	lower turnout
block5_ksi_collision_events_2021_2025_per_1000_x_block1_unemployment_rate_share	0.767	-0.213	lower turnout
block1_average_household_size_x_block1_age_35_64_share	0.753	0.210	higher turnout
block1_average_household_size_x_block5_ksi_collision_events_2021_2025_per_1000	0.725	-0.202	lower turnout
block1_low_income_lim_at_share_x_block5_requests_311_per_1000	0.704	0.196	higher turnout
block1_age_65_plus_share_x_block1_average_household_size	0.572	-0.159	lower turnout

Interpretation Of Retained Interactions

- bachelor or higher education share with effective mayoral candidates above five percent: this asks whether education or class resources change how local electoral fragmentation relates to turnout. It is useful for a story about whether complex local contests are easier to navigate in higher-resource places.
- KSI collisions per 1000 residents with non official mother tongue share: this asks whether service-contact geography has a different meaning in immigrant, language, or citizenship contexts. It may point to places where municipal service contact and civic participation are linked unevenly.
- low income share with age 35 to 64 share: this asks whether class or household structure has a different turnout relationship depending on the age profile of the tract. It helps separate youth, working-age, and older-adult versions of the turnout story.
- KSI collisions per 1000 residents with unemployment rate: this asks whether service-contact intensity is associated with turnout differently in higher-need or higher-resource places. It is especially relevant for interpreting 311 requests, development applications, and collision exposure as civic-contact context rather than simple services.
- average household size with age 35 to 64 share: this asks whether class or household structure has a different turnout relationship depending on the age profile of the tract. It helps separate youth, working-age, and older-adult versions of the turnout story.
- average household size with KSI collisions per 1000 residents: this asks whether service-contact intensity is associated with turnout differently in higher-need or higher-resource places. It is especially relevant for interpreting 311 requests, development applications, and collision exposure as civic-contact context rather than simple services.
- low income share with 311 requests per 1000 residents: this asks whether service-contact intensity is associated with turnout differently in higher-need or higher-resource places. It is especially relevant for interpreting 311 requests, development applications, and collision exposure as civic-contact context rather than simple services.
- age 65 plus share with average household size: this asks whether class or household structure has a different turnout relationship depending on the age profile of the tract. It helps separate youth, working-age, and older-adult versions of the turnout story.

Reference Categories For Interactions

The interaction results mainly point to four conditional reference categories: class and service contact, education and local electoral fragmentation, language or immigrant geography and service contact, and age structure interacting with class or household size. These should not all become final hypotheses automatically. They are candidate story lines that can be checked against maps, plots, and substantive plausibility.

Remaining VIF In Interaction Model

Variable	Family	VIF
block1_age_18_34_share	age_structure	118.541
block1_age_65_plus_share	age_structure	103.655
block5_school_age_5_17_share	service_contact	45.526
block1_age_35_64_share	age_structure	38.511
block1_average_household_size	household_class	26.825
block5_ksi_collision_events_2021_2025_per_1000__ x_block1_unemployment_rate_share	other	16.688
block1_low_income_lim_at_share__x_block5_reque sts_311_per_1000	other	16.204
block2_apartment_share	housing_form_density	15.919
block5_ksi_collision_events_2021_2025_per_1000__ x_block3_non_official_mother_tongue_share	other	15.433
block2_renter_share	housing_tenure	15.194
block3_non_citizen_share	immigration_citizenship	14.031
block1_average_household_size__x_block5_ksi_coll ision_events_2021_2025_per_1000	other	11.071
block2_condo_share	housing_form_density	10.652
block2_same_address_5yr_share	residential_stability	10.401
block3_recent_immigrant_share	immigration_citizenship	10.269

Interpretation

Interactions should be used as story devices only when they describe a plausible conditional relationship. The strongest candidates here are not arbitrary products; they mainly ask whether the turnout penalty associated with social composition becomes sharper in places with particular housing form, electoral fragmentation, or service/contact contexts. If an interaction improves prediction only trivially, it should still be kept out of the final narrative unless it clarifies a substantive mechanism.